1) Here are the first five terms of a sequence.

$$2 \quad 4 \quad 7 \quad 11 \quad 16$$

Write down the next two terms in the sequence.

2) The first term in a sequence is 3. The term to term rule is add 5.
Is 97 a term in the sequence? Give a reason for your answer.

3) Here are the first five terms of a Fibonacci sequence

$$1 \quad 2 \quad 3 \quad 5 \quad 8$$

Write down the next two terms in the sequence.

4) The nth term of a sequence is $4n + 3$
   a) Find the first two terms of this sequence.
   b) Is 35 a term in this sequence. You must show how you get your answer.

5) The nth term of a sequence is $n^2 + 1$
   a) Find the first two terms of this sequence.
   b) Is 35 a term in this sequence. You must show how you get your answer.

6) Here are the first 5 terms of a sequence.

$$17 \quad 14 \quad 11 \quad 8 \quad 5$$

   a) Find the next term of this sequence.

   The nth term of a different sequence is $10n^2 + 5$

   b) Work out the 5th term of this sequence.

7) Here are the first four terms of a sequence.

$$7 \quad 13 \quad 19 \quad 25$$

   a) Write down the next term in the sequence.
   b) Explain how you got your answer

8) Here are the first four terms of a number sequence.

$$2 \quad 3 \quad 5 \quad 9$$

The rule to continue the sequence is multiply the previous term by 2 and then subtract 1.
Work out the 5th term of this sequence.

9) Here are the first 5 terms of a Fibonacci sequence.

$$2 \quad 2 \quad 4 \quad 6 \quad 10$$

Find the 8th term of this sequence.

10) The nth term of a sequence is $n^2 + 3$.
   a) Find the first three terms of this sequence.
   b) Find the 10th term in this sequence.

11) Here are the first five terms of a sequence.

$$31 \quad 27 \quad 23 \quad 19 \quad 15$$

a) Find the first negative term in the sequence.
b) Is –30 a term in this sequence? Give a reason for your answer.

12) Here are the first 5 terms of an arithmetic sequence.

$$-3 \quad 1 \quad 5 \quad 9 \quad 13$$

a) Find an expression, in terms of n, for the nth term of this sequence.

The nth term of a different arithmetic sequence is $2n - 3$

b) Is 101 a term in this sequence? Show how you get your answer.

13) Here are the first 5 terms of a sequence.

$$9 \quad 14 \quad 19 \quad 24 \quad 29$$

Find an expression, in terms of n, for the nth term of this sequence.

14) Here are the first 5 terms of a sequence.

$$25 \quad 22 \quad 19 \quad 16 \quad 13$$

Find an expression, in terms of n, for the nth term of this sequence.

15) Here are the first four terms of an arithmetic sequence.

$$4 \quad 11 \quad 18 \quad 25$$

Write down an expression, in terms of n, for the nth term of the sequence.

16) Here are the first four terms of an arithmetic sequence.

$$35 \quad 31 \quad 27 \quad 23$$

Write down an expression, in terms of n, for the nth term of the sequence.

17) Here are the first five terms of an arithmetic sequence.

$$21 \quad 27 \quad 33 \quad 39 \quad 45$$

Write down an expression, in terms of n, for the nth term of the sequence.

18) Here are the first five terms of an arithmetic sequence

$$2 \quad 7 \quad 12 \quad 17 \quad 22$$

Write down an expression, in terms of n, for the nth term of the sequence.

19) Find the nth term of the following quadratic sequences:
a) $-8, 2, 16, 34 \ldots$
b) $6, 10, 16, 24 \ldots$
c) $-2, -1, 1, 4 \ldots$
d) $19, 15, 9, 1 \ldots$
e) $2, 10, 24, 44 \ldots$
f) $15, 19, 25, 33 \ldots$
g) $2, 10, 22, 38 \ldots$

20) Write down the next two terms in the following quadratic sequence

$$9, 13, 19, 27 \ldots$$